

INDU 211: Introduction to Production & Manufacturing Systems

Problem Solutions Video Series — Complete Roadmap & Curriculum Guide

Department of Mechanical, Industrial & Aerospace Engineering · Concordia University

1. Overview of the Video Tutorial Series

This master guide accompanies the official **INDU 211 Problem Solutions** video tutorial series for Concordia University. The series provides comprehensive walkthroughs of seven core quantitative problem types that appear on Concordia midterm and final examinations.

- **Full YouTube Playlist:** [INDU 211 - Problem Solutions Playlist](#)
- **Total Duration:** ~1 hour 56 minutes of high-yield quantitative problem solving
- **Course Alignment:** Direct pedagogical correlation with Dr. Masoumeh Kazemi Zanjani's lecture curriculum and Turner, Mize, Case & Nazemetz's

Introduction to Industrial and Systems Engineering

(3rd Edition).

2. Curriculum Decoupling: Midterm vs. Final Exam Scope

To optimize your study strategy, the 7 problem types are strictly decoupled below into the **Midterm Exam (Week 6, 35% of Grade)** and the **Final Exam (Weeks 7–12, 50% of Grade)**:

Guide #	Problem Title & Video Link	Duration	Curriculum Chapter	Target Exam	Core Method / Formula
01	Demand Forecasting	20:35	Chapter 7: Operations Planning & Control	 MIDTERM	n -Period Moving Average: $\hat{X}_t = \frac{1}{n} \sum_{i=1}^n X_{t-i}$, Forecast Error , Market Share $W = \frac{\text{Sales}}{\text{Share}}$
05	Warehouse TSP Routing	15:24	Chapter 5: Materials Handling & Routing	 MIDTERM	2020 Midterm Exam Problem 2: Nearest Neighbor Heuristic, Sub-optimality Proof, Tour Permutations $(n - 1)!$
02	Linear Programming Minimization	19:44	Chapter 14: Deterministic OR Models	🔍 FINAL	Graphical Method with Equality Constraint ($X_1 + 3X_2 = 9$), 1D Line-Segment Feasible Space, Iso-Cost Propagation
06	LP Production Modeling	18:11	Chapter 14: Deterministic OR Models	🔍 FINAL	Multi-Product Drug Formulation: Profit Objective, Raw Material Inequalities, Machine Time Equivalence, 2 : 1 : 3 Ratio
04	Queuing Theory (M/M/1)	17:23	Chapter 15: Probabilistic OR Models	🔍 FINAL	Poisson Arrivals λ , Service Rate μ , Traffic Intensity $\rho = \frac{\lambda}{\mu}$, $W_q = \frac{\lambda}{\mu(\mu - \lambda)}$, $P(n \geq 2) = \rho^2$
07	Statistical Quality Control	18:10	Chapter 8: Quality Control & SPC	🔍 FINAL	$\bar{X} - R$ Charts, Out-of-Control Sample Elimination, Process Standard Deviation $\hat{\sigma} = \frac{\bar{R}}{d_2}$, Capability $C_p = \frac{USL - LSL}{6\hat{\sigma}}$
03	PERT / CPM Project Scheduling	6:52	Chapter 17: Project Management	🔍 FINAL	Activity-on-Arrow (AOA) Network, Path Durations, Critical Path Identification (15 Weeks), Total Activity Slack

3. Master Formula Quick Reference

Module 1: Forecasting (n -Period Moving Average)

$\hat{X}_t$

$$t = \frac{1}{n} \sum_{i=1}^n X_{t-i}$$

$$X_{t-1} + X_{t-2} + \dots + X_{t-n}$$

$e_t = X_t - \hat{X}_t \quad (\text{Positive } e_t \implies \text{Underestimation / Stockout})$

$$W = \frac{\text{Company Sales}}{\text{Company Market Share Percentage}}$$

Module 2: Traveling Salesperson Problem & Search Space

Total Closed Tours from Depot: $N_{\text{routes}} = (n - 1)!$

Nearest Neighbor Rule: From current node i , pick unvisited j with $\min(d_{ij})$

Module 3: Linear Programming Graphical Corner Evaluation

Objective Function: $\min Z = c_1X_1 + c_2X_2$

Corner Point Search: Evaluate Z at all extreme boundary vertices; select minimum

Module 4: $M/M/1$ Standard Queuing Formulas

Utilization Factor: $\rho = \frac{\lambda}{\mu}$ (Stable if $\rho < 1$)

Expected Waiting Time in Queue: $W_q = \frac{\lambda}{\mu(\mu - \lambda)}$

Expected Total Time in System: $W = W_q + \frac{1}{\mu} = \frac{1}{\mu - \lambda}$

Probability of Exactly n Customers: $P_n = (1 - \rho)\rho^n$

Probability of $\geq k$ Customers: $P(n \geq k) = \rho^k$

Module 5: Statistical Process Control & Process Capability

$\bar{\bar{X}}$ (Revised Grand Mean: $\bar{\bar{X}}$)

$$\bar{\bar{X}} = \frac{\sum \bar{X}}{m}$$

$\bar{X}_{\text{initial}} - \sum \bar{X}_{\text{out-of-control}}$ (m - k)

$\hat{\sigma}$ (Estimated Process Standard Deviation: $\hat{\sigma} = \frac{\bar{R}}{d_2}$)

$C_p = \frac{\text{USL} - \text{LSL}}{6\hat{\sigma}}$ (C_p ≥ 1.33 implies Capable)

Module 6: Project Scheduling (PERT / CPM)

Critical Path: Sequence of dependent activities with the longest cumulative duration

Total Slack for Activity i : $\text{Slack}_i = \text{LS}_i - \text{ES}_i = \text{LF}_i - \text{EF}_i$